AI Use Cases in Air Transportation

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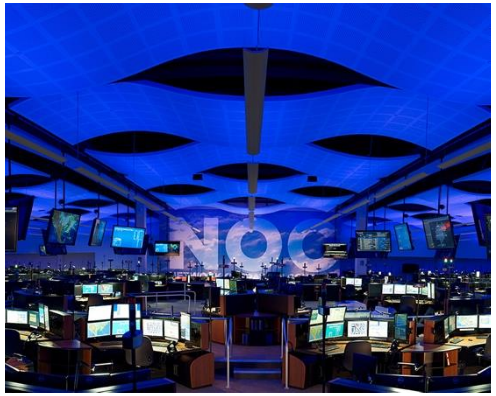
Intelligent Aerospace Systems Lab (IASL)

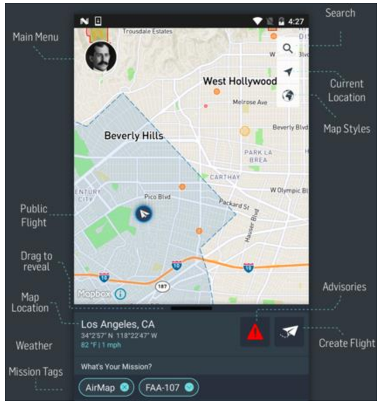
Models and algorithms for design and operation of air transportation and aviation systems

- **Methods:** control, optimization, machine learning, artificial intelligence
- **Applications:**

Air Traffic Control/Management



Airline Operations



UAS Traffic Management

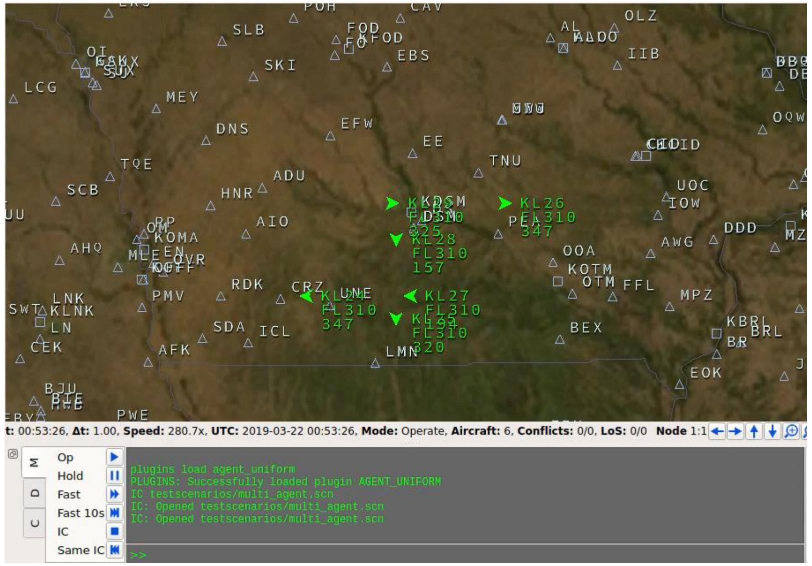
Vehicle Battery Prognostics

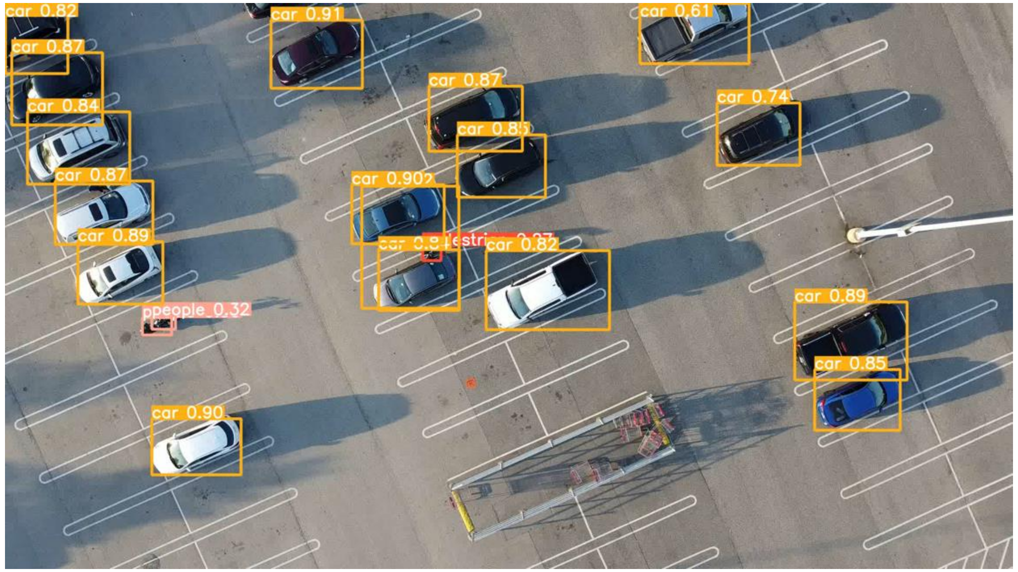
eVTOL Urban Air Mobility

Vertiport Operations

Current Research Focuses

- Aircraft autonomy, multi-agent autonomy and human-autonomy teaming
- Aviation operations in ATC/M, UAM, UTM and airports/vertiports
- AI safety and certification in aviation systems





Use Case I: Aircraft Separation Assurance (2018 to today)

Research Motivation

- In the controlled airspace, human ATCs will benefit from an effective AutoATC tool
- In future high-density UTM and AAM, human operators or human-based ATC services will not scale

Related Work – Collision Avoidance and Conflict Resolution

- **Traffic Alert and Collision Avoidance System (TCAS)**
  - Air to air, one on one
  - Last defense against mid air collision
  - 15 to 35 seconds before crash
  - Vertical maneuvers
- **ACAS X**
  - Same display
  - Same hardware
  - Different advisory logic (fixed rules vs. probabilistic model)
  - Solving a MDP with discrete state space
- **Autoresolver and T-SAFE**
  - Ground based
  - Conflict resolution (prevent LOS)
  - 35 seconds to 8 mins before crash

Multi-Agent Reinforcement Learning Formulation

- **State:** information of the ownship, and its k closest aircraft

Ownship	Intruder 1	Intruder 2	Intruder 3
d_g^o	d_g^{i1}	d_g^{i2}	d_g^{i3}
v^o	v^{i1}	v^{i2}	v^{i3}
$\dot{v}^o$	$\dot{v}^{i1}$	$\dot{v}^{i2}$	$\dot{v}^{i3}$
d_m^o	d_m^{i1}	d_m^{i2}	d_m^{i3}
p^o	p^{i1}	p^{i2}	p^{i3}
r^o	r^{i1}	r^{i2}	r^{i3}
$\boldsymbol{s}^o$	$\boldsymbol{s}^{\mathrm{i}}$		

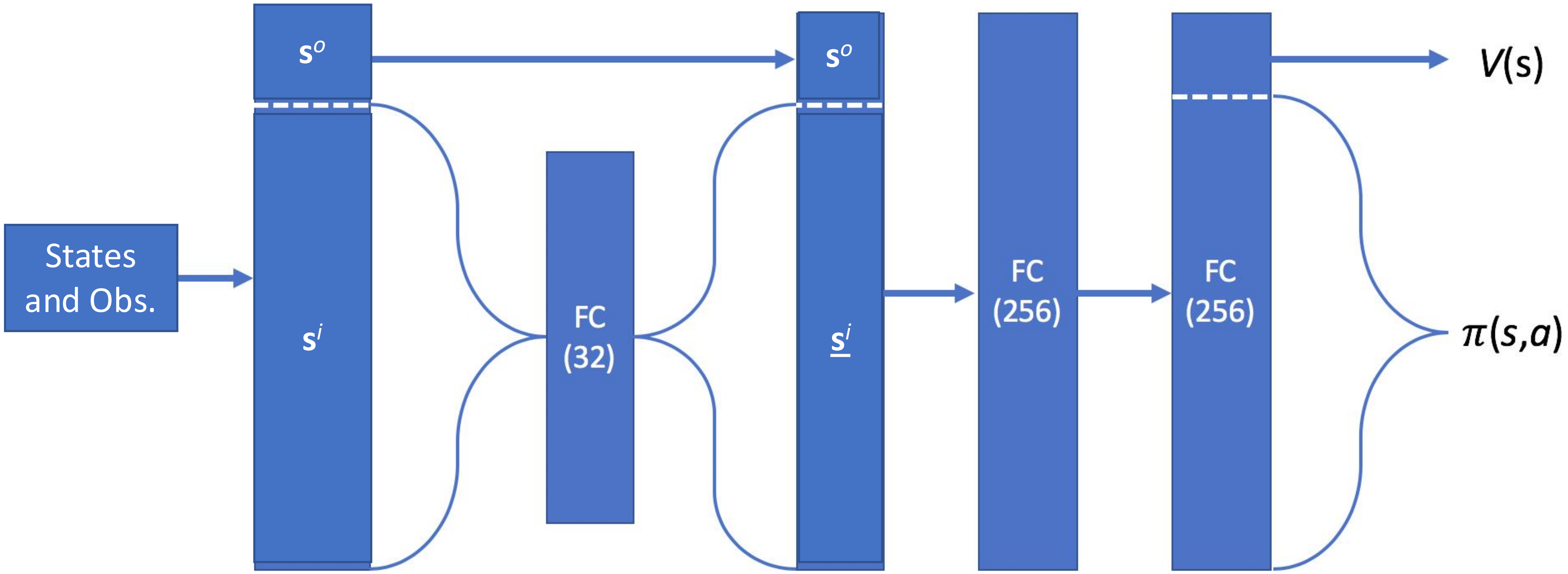
- **Action:** each aircraft can speed up, slow down, or hold the current speed

$$A = \{v_-, v_{hold}, v_+\}$$

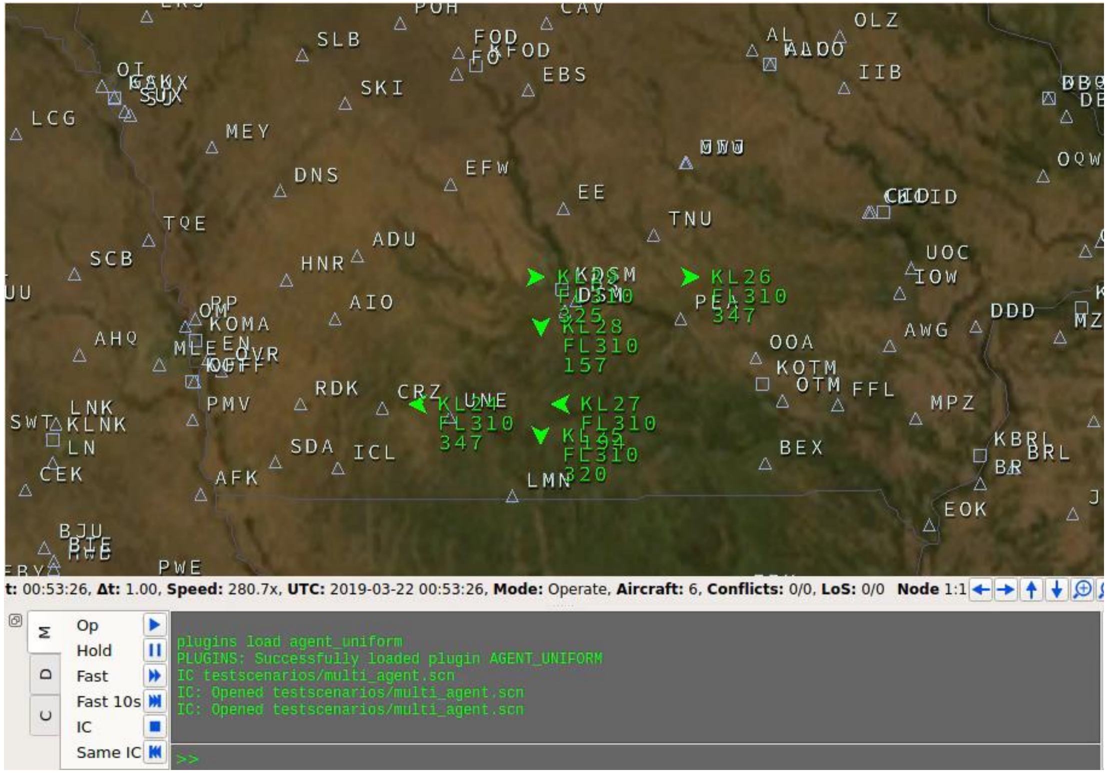
- **Reward:** we penalize each LOS event and prevent two aircraft get too close

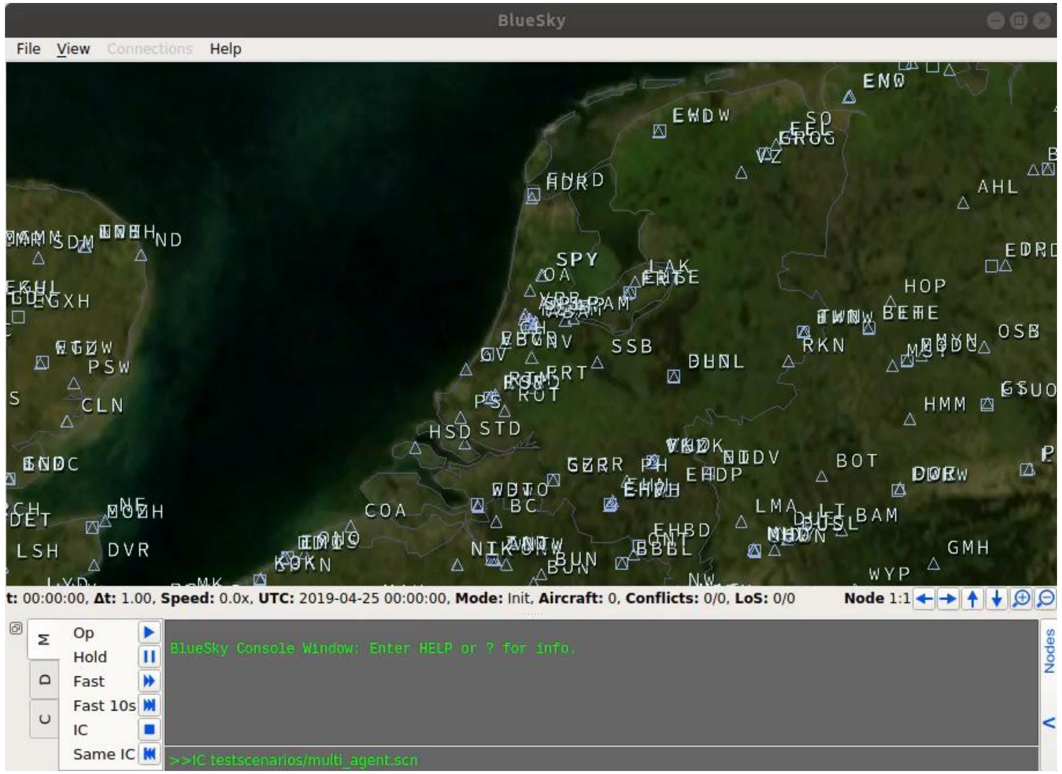
$$r = \begin{cases} -1 & (\leq 3\ NM) \\ \varepsilon & (\text{in between}) \\ 0 & (> 10\ NM) \end{cases}$$

Deep Learning Architecture of MARL

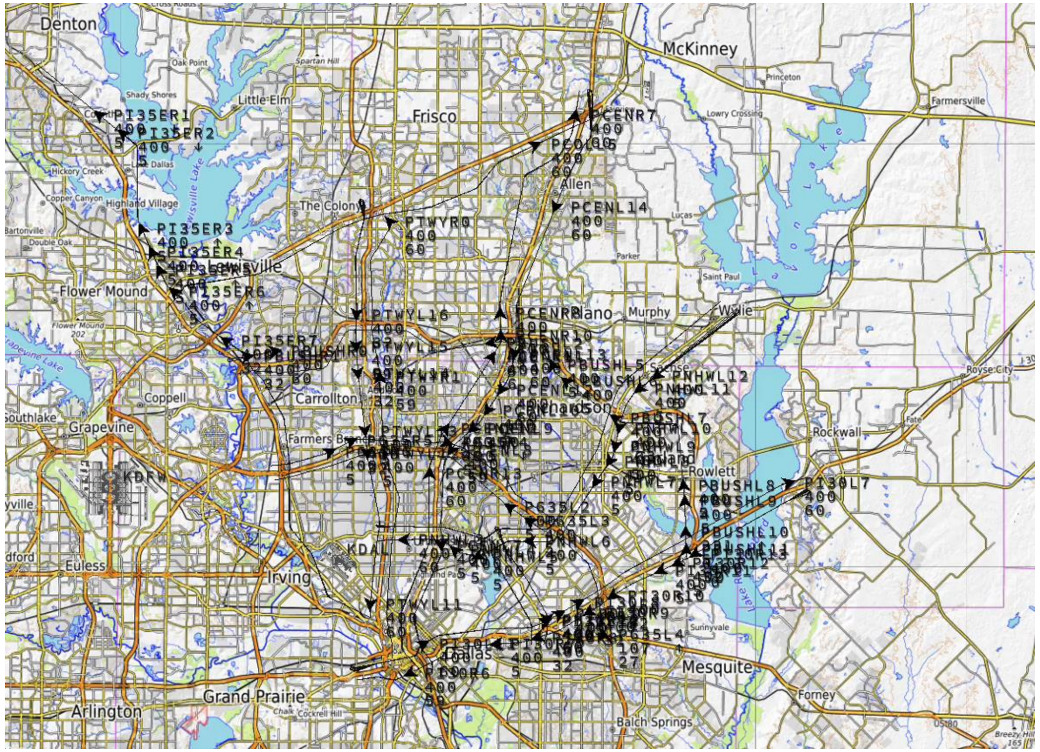


Intersections and Merging Fix in Structured Airspace





Validations of the MARL based AutoATC



Dallas Fort Worth Large-scale Simulation
Fall, 2023

Flight Test at MIT Lincoln Lab
Summer, 2024

Use Case II: Vision-based AutoLand
(2020 to today)

Research Motivation

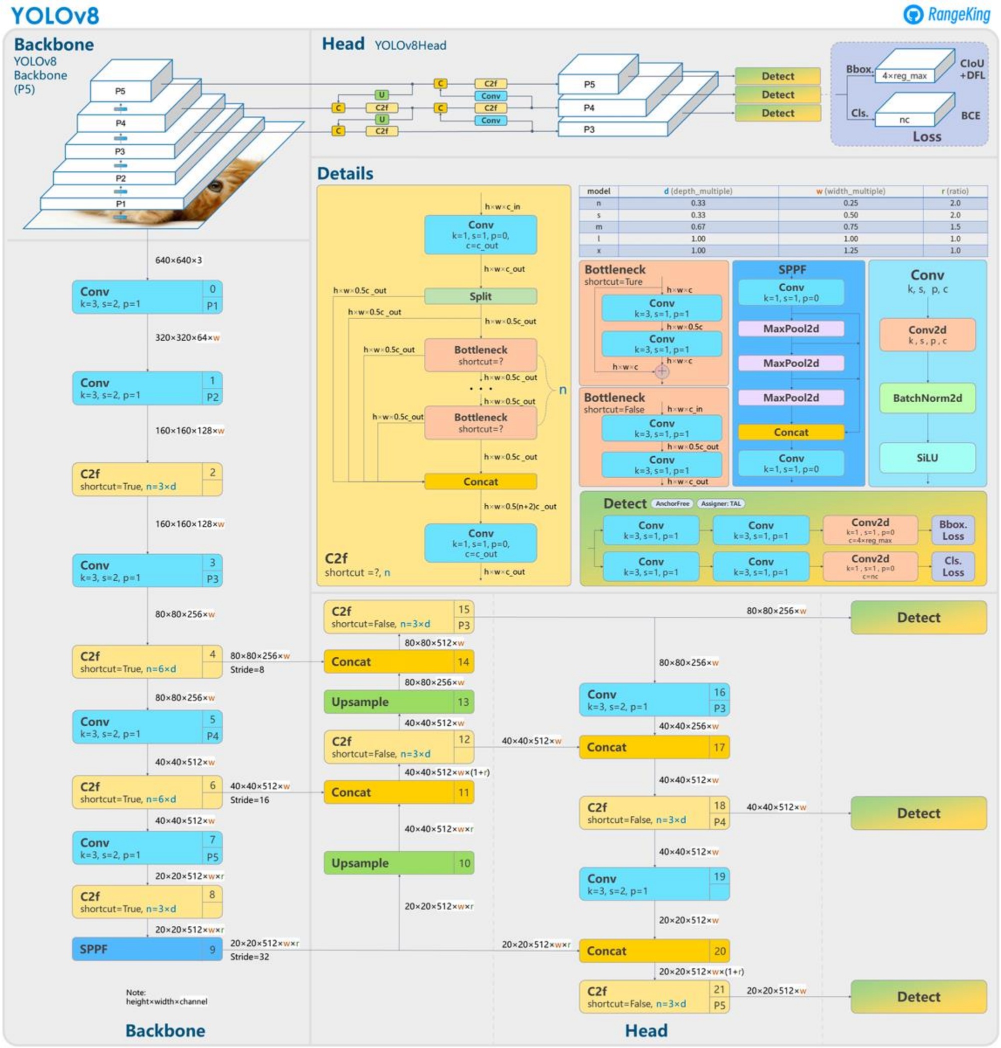
- Autonomous landing or dropping for drone delivery (Wing, Prime Air, and Zipline)
- Can we identify the objects on/near the landing pad using a computer vision model?

Photo Credit: Wing

Highlights of the DNN based Vision Model

- YOLOv8
- Pre-trained on COCO dataset
- Fine tuned with VisDrone dataset and annotated dataset
- Landing pad detection from ArUco markers
- Object detection with YOLO
- Landing pad occupancy detection algorithm with bounding boxes
- Trimmed YOLO on onboard GPU Jetson Xavier NX

YOLOv8 Architecture



Credit: roboflow.com

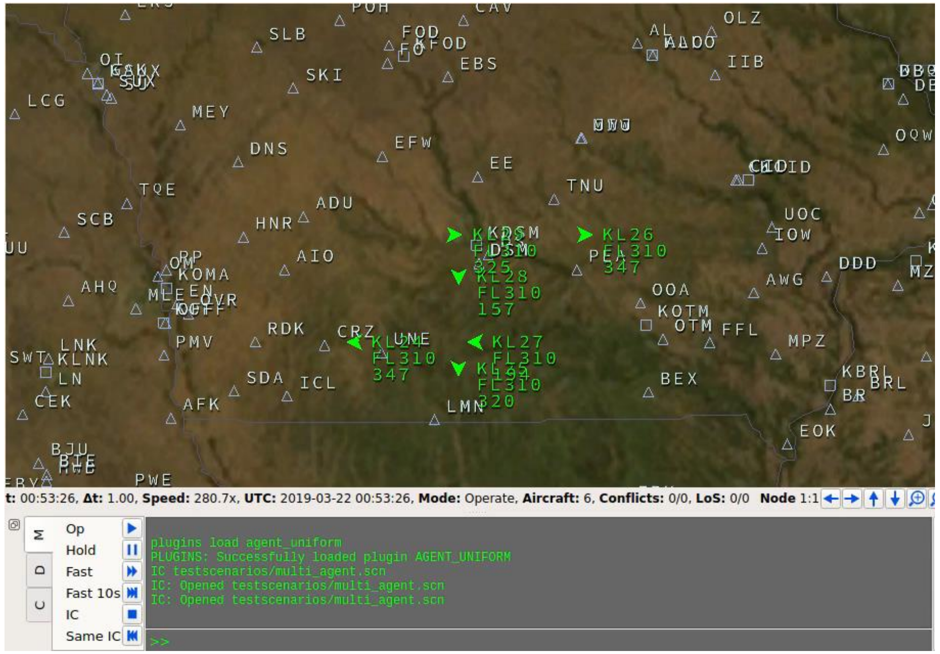
Landing Pad Occupancy Detection

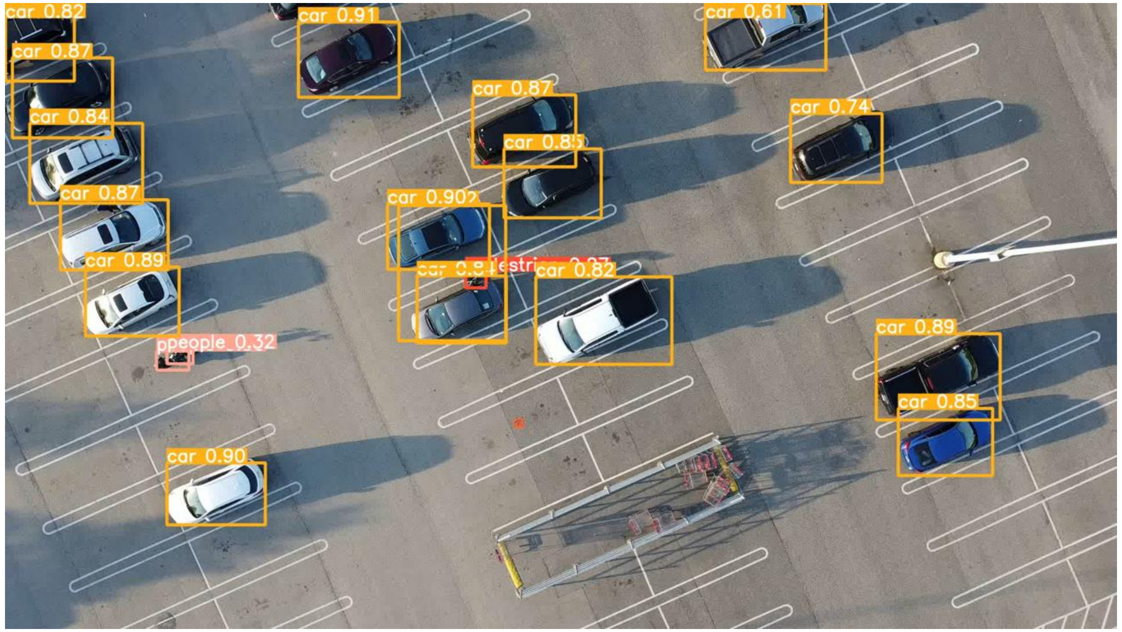
Landing Pad Occupancy Detection

Landing Pad Occupancy Detection in Simulation

AI Safety and Verification Efforts

Two Open-source AI Use Cases for Aviation Applications





Neural Net for Control
Safety Critical

Neural Net for Perception
Safety Critical

Ongoing Verification Efforts

- Verify the dataset sufficiency and coverage completeness
- Study the mathematical/statistical bounds of the neural nets, instead of treating it as a black box
- Design-time formal verification
- Run-time safety monitoring and shielding
- Adaptive stress testing and falsification
- Explainability and human-AI teaming

Questions and comments

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